Implemented memory-initialized LUT system by loading HEX file (test.hex) into instruction memory (IMEM).

Supported instruction set:

- RRR

OC	R1	R2	R3	Func		Assembly-Code Format	Meaning
3	3	3	3	4			
000	Rs1	Rs2	Rd	0000	add	add Rs1, Rs2, Rd	Reg[Rd] ← Reg[Rs1] + Reg[Rs2]
000	Rs1	Rs2	Rd	0001	sub	sub Rs1, Rs2, Rd	Reg[Rd] ← Reg[Rs1] - Reg[Rs2]
000	Rs1	000	Rd	0010	inc	inc Rs1, Rd	Reg[Rd] ← Reg[Rs1] + 1
000	Rs1	000	Rd	0011	dec	dec Rs1, Rd	Reg[Rd] ← Reg[Rs1] - 1
000	Rs1	Rs2	Rd	0100	and	and Rs1, Rs2, Rd	Reg[Rd] ← Reg[Rs1] and Reg[Rs2]
000	Rs1	Rs2	Rd	0101	or	or Rs1, Rs2, Rd	Reg[Rd] ← Reg[Rs1] or Reg[Rs2]
000	Rs1	Rs2	Rd	0110	xor	xor Rs1, Rs2, Rd	Reg[Rd] ← Reg[Rs1] xor Reg[Rs2]
000	Rs1	Rs2	Rd	0111	nand	nand Rs1, Rs2, Rd	Reg[Rd] ← Reg[Rs1] nand Reg[Rs2]
000	Rs1	000	Rd	1000	shfl	shfl Rs1, Rd	Reg[Rd] ← Reg[Rs1] << 1
000	Rs1	000	Rd	1001	shfr	shfr Rs1, Rd	Reg[Rd] ← Reg[Rs1] >> 1
000	Rs1	000	Rd	1010	shll	shll Rs1, Rd	Reg[Rd] ← Reg[Rs1] << 2
000	Rs1	000	Rd	1011	shrr	shrr Rs1, Rd	Reg[Rd] ← Reg[Rs1] >> 2
000	Rs1	000	Rd	1100	slll	slll Rs1, Rd	Reg[Rd] ← Reg[Rs1] << 3
000	Rs1	000	Rd	1101	srrr	srrr Rs1, Rd	Reg[Rd] ← Reg[Rs1] >> 3
000	Rs1	Rs2	Rd	1110	(mul)	mul Rs1, Rs2, Rd	Reg[Rd] ← Reg[Rs1] x Reg[Rs2]
000	Rs1	Rs2	Rd	1111	(div)	div Rs1, Rs2, Rd	Reg[Rd] ← Reg[Rs1] ÷ Reg[Rs2]

- RRI

OC	R1	R2	Imm		Assembly-Code Format	Meaning
3	3	3	7			
001	Rs1	Rd	Imm	addi	addi Rs1, Rd, Imm	Reg[Rd] ← Reg[Rs1] + Imm
010	Rs1	Rd	Imm	subi	subi Rs1, Rd, Imm	Reg[Rd] ← Reg[Rs1] – Imm
011	Rs1	Rs2	Imm	beq	beq Rs1, Rs2, Imm	PC = Reg[Rs1] == Reg[Rs2] ? Imm : PC + 1

- RI

OC	R1	Imm		Assembly-Code Format	Meaning
3	3	9			
100	Rd	Imm	lw	lw Rd, Imm	Reg[Rd] ← Mem[Imm]
101	Rd	Imm	sw	sw Rd, Imm	Reg[Rd] → Mem[Imm]
110	Rd	Imm	jpnz	jpnz Rd, Imm	PC = Reg[Rd] != 0 ? Imm : PC + 1
111	Rd	Imm	li	li Rd, Imm	Reg[Rd] ← Imm

- Extend

001	Rs1	Rs1	0000000		nop	nop	(addi)
000	Rs1	Rs1	Rd	0111	not	not Rs1, Rd	(nand)
101	Rs1	Rs1	Imm		jmp	jmp Imm	(beq)
000	Rs1	Rs1	Rd	0100	mov	mov Rs1, Rd	(and)